

Viet Minh Hieu Nguyen

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EDUCATION

Union College

Bachelor of Science in Computer Science and Mathematics, Minor in Statistics

Schenectady, NY

Sep 2024 – Jun 2028

- **GPA:** 3.97 | **Honors:** Dean's List
- **Relevant Coursework:** Object-Oriented Programming, Data Structures and Algorithms, Linear Algebra, Mathematical Statistics, Probability Theory, Abstract Algebra, Discrete Mathematics, Multivariable Calculus
- **Activities:** ICPC Team Member, ACM Member, IEEE Member

RELATED EXPERIENCE

AlgoVerse

Machine Learning Researcher

Remote

Nov 2025 – Present

- Improved GSM8K accuracy by 20% over the CODI baseline by discovering that a specific bracket notation around explicit reasoning steps significantly improves latent thought conditioning.
- Built a logit lens analysis tool to characterize how answer's information propagates through CODI's latent representations, decoding residual hidden states into layer-by-position probability heatmaps.
- Designed controlled ablation studies over CODI's latent reasoning positions, using configurable probing and structured wrong-answer filtering to pinpoint where and why implicit reasoning breaks down

Union College CS Department

Undergraduate Researcher

Schenectady, NY

Oct 2025 – Present

- Developed a Tensor class in Python using Numpy to implement Strassen's Algorithm, achieving a 12.5% reduction in multiplication operations for $n \times n$ matrices by lowering time complexity from $O(n^3)$ to $O(n^{2.81})$
- Investigated algorithmic discovery methods by analyzing SAT-based local search techniques used to find new 3×3 schemes, comparing the efficiency of Laderman's 23-multiplication method against standard recursive models.

Union College

Information Technology Support Specialist

Schenectady, NY

Jan 2025 – Sep 2025

- Developed and maintained a repository of PowerShell and Bash scripts to automate repetitive diagnostic tasks, reducing manual troubleshooting time by 23% and ensured consistent data collection for escalations
- Drove faster triage and consistent escalations with standardized runbooks as the first point of contact for technical assistance for all professors and students

PROJECTS

BioRT-Bench | [GitHub](#) | [Writeup](#)

Apr 2026

- Designed BioRT-Bench with a team of 4 at the Apart Research Hackathon (2nd place, AI Biosecurity Tools track): the first multi-model multi-attack red-teaming benchmark for bio-misuse safeguards in frontier LLMs.
- Built a calibrated bio-aware LLM judge extending StrongREJECT with two orthogonal dimensions (specificity, actionability) to score model responses on refusal robustness.
- Implemented the red-teaming harness in Microsoft PyRIT with four attack methods (direct, base64, PAIR, Crescendo); ran 640 scored attacks across 4 production models and 5 biosecurity categories.

Predicting LLM Chain-of-Thought Robustness | [GitHub](#) | [Writeup](#)

Jan 2026

- Predicted chain-of-thought resilience with 75.4% accuracy by training L2-regularized linear probes on Qwen3-4B's residual stream activations, replacing the standard 20-sample resampling approach with a single forward pass
- Identified Layer 13 as the peak resilience encoding layer with 0.793 test AUC by extracting and probing activation vectors across all 36 layers of Qwen3-4B over 7,000 reasoning sentences using TransformerLens
- Recovered a interpretable resilience direction in activation space with $r=0.42$ correlation to ground-truth scores by projecting logistic regression weights back through PCA into the original 2560-dimensional residual stream

- Engineered an NLP preprocessing pipeline using NLTK and Jupyter Notebooks, implementing tokenization and lemmatization to clean body text from diverse datasets.
- Optimized dataset integrity by executing a balanced undersampling strategy on 50,000 entries to mitigate class imbalance and feature bias.
- Evaluated 5 ML models (LightGBM, SVM, RF, NB, LR), achieving a peak 97.7% accuracy and 0.98 F1-score with LightGBM to detect sophisticated social engineering tactics

- Developed a full-stack productivity app using React, Express and Postgres, that guides 14 concurrent users, leading to over 200+ focused work sessions using the Pomodoro technique
- Engineered an AI productivity coach using Express and the OpenAI GPT-4o API, delivering personalized focus strategies to users through a CI/CD pipeline on Vercel.

LEADERSHIP & EXTRACURRICULARS

Union College Robotics Club

Treasurer

Schenectady, NY

Nov 2024 – Present

- Collaborated in a team of four to build a maze-solving robot using the Flood Fill algorithm in C++
- Secured 5th place out of 20 teams at the 2024 IEEE Region 1 Micromouse Competition at MIT
- Elected as a treasurer by membership to manage the club's annual budget of \$3000

OTHER EXPERIENCE

Union College

Chemistry Tutor

Schenectady, NY

Jan 2025 – Mar 2025

- Improved 150+ undergraduate students' average grades and understanding general chemistry by identifying knowledge gaps and fostering critical-thinking skills through problem-based learning

SKILLS

Programming Languages: Python, SQL, JavaScript/TypeScript, HTML/CSS, C/C++, Java, Bash/Power

Frameworks: NumPy, pandas, PyTorch, Matplotlib, Seaborn, React, Express.js, Node.js, TailwindCSS

Tools & Libraries: Git/GitHub, Huggingface, Postgres, Docker, SupaBase, Arduino, Sci-kit Learn, Linux, Excel